QQ-A-250/7E
November 23, 1970
SUPERSEDING
QQ-A-250/7D
17 March 1967
(See 6.3)

FEDERAL SPECIFICATION

ALUMINUM ALLOY 5086, PLATE AND SHEET

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

(This specification forms a part of the latest issue of Federal Specification QQ-A-250.)

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for aluminum alloy 5086 plate and sheet; the general requirements are covered in QQ-A-250.

1.2 Classification.

1.2.1 Tempers. The plate and sheet shall be classified in one of the following tempers, as specified (see 6.2).

O - Annealed.
H32 - Strain-hardened one-quarter hard temper and then stabilized.
H34 - Strain-hardened half-hard temper and then stabilized.
H36 - Strain-hardened three-quarters hard temper and then stabilized.
H38 - Strain-hardened full-hard temper and then stabilized.
H112 - As rolled, properties as specified in table II.

2. APPLICABLE DOCUMENTS

Latest issue of Federal Specification:

QQ-A-250 - Aluminum and Aluminum Alloy Plate and Sheet; General Specification for.

FSC 9535

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3. REQUIREMENTS

3.1 Chemical composition.

3.1.1 The chemical composition shall conform to the requirements specified in table I.

TABLE I. Chemical composition[1/]

Element	Percent	
	Minimum	Maximum
Magnesium	3.5	4.5
Manganese	0.20	0.7
Chromium	.05	.25
Iron	-	.50
Silicon	-	.40
Zinc	-	.25
Titanium	-	.15
Copper	-	.10
Others, each	-	.05
Others, total	-	.15
Aluminum	Remainder	

1/ Analysis shall regularly be made only for the elements specifically mentioned in the table. If, however, the presence of other elements is indicated in the course of routine analysis further analysis shall be made to determine conformance to the limits specified for other elements.

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3.2 Mechanical properties.

3.2.1 Mechanical properties of material as supplied. The mechanical properties parallel to the direction of final rolling shall conform to the requirements of table II for the temper specified.

TABLE II. Mechanical properties

Temper	Thickness	Tensile strength		Yield strength 1/, minimum	Elongation in 2 in. or 4 times D 2/, 3/, minimum
		Minimum	Maximum		
	Inches	P.s.i.	P.s.i.	P.s.i.	Percent
0	0.020 thru 0.050	35,000	44,000	14,000	15
	.051 thru .249	35,000	44,000	14,000	18
	.250 thru 2.000	35,000	44,000	14,000	16
H112	0.188 thru 0.499	36,000	-	18,000	8
	.500 thru 1.000	35,000	-	16,000	10
	1.001 thru 2.000	35,000	-	14,000	14
	2.001 thru 3.000	34,000	-	14,000	14
H32	0.020 thru 0.050	40,000	47,000	28,000	6
	.051 thru .249	40,000	47,000	28,000	8
	.250 thru 2.000	40,000	47,000	28,000	12
H34	0.009 thru 0.019	44,000	51,000	34,000	4
	.020 thru .050	44,000	51,000	34,000	5
	.051 thru .249	44,000	51,000	34,000	6
	.250 thru 1.000	44,000	51,000	34,000	10
H36	.006 thru 0.019	47,000	54,000	38,000	3
	.020 thru .050	47,000	54,000	38,000	4
	.051 thru .162	47,000	54,000	38,000	6
H38	0.006 thru 0.020	50,000	-	41,000	3

1/ At 0.2 percent offset.

2/ Not required for material 1/2 inch or less in width.

3/ D represents specimen diameter.

4. QUALITY ASSURANCE PROVISIONS
(See QQ-A-250.)

5. PREPARATION FOR DELIVERY
(See QQ-A-250.)

6. NOTES

6.1 Intended use. This alloy is for use where a weldable moderate strength alloy having good corrosion resistance is required.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

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(a) Title, number, and date of this specification.

(b) Form and temper required (see 1.2.1).

(c) Dimensions required.

(d) Requirements for sizes not specifically covered (see QQ-A-250).

(e) Selection of applicable levels of preservation, packaging, and packing required, if other than level C (see QQ-A-250).

6.3 Certain provisions of this specification are the subject of international standardization agreement ABC-NAVY-STD-44. When amendment, revision, or cancellation of this specification is proposed which will effect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization officer, if required.

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SPECIFICATION

QQ-A-250/7E ALUMINUM ALLOY 5086, PLATE AND SHEET

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

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